Contrast (PD / PAX / both) — run report

RUN

011348_pd-visibility_rp-quick-test
Status: failed | Started: 2026-10-07T00:13:48.041289+00:00

PURPOSE

Measure PD voltage contrast, PAX power contrast, or both concurrently. Passive mode leaves the drive untouched; active mode drives the selected phase actuator and returns outputs to zero on exit.

REQUIRED OPTICAL SETUP

PD needs a DC-coupled input and measured/provided signed dark baseline. PAX needs the recorded port/wavelength. Both requires light delivered to both sensors (e.g. a beam split), with that arrangement recorded. Active mapping: phi1=OUT1, phi2=OUT2; the other output is zero. Passive PD needs the existing Pyrpl FPGA. Drive must span fringes.

RECORDED BENCH CONTEXT (RECIPE)

Source: pd; mode: passive; PD input: in1; PAX location: In front of NPBS 1, C port; requested/declared drive: 0.5 Hz. RP outputs unchanged.

RUN COMMENT

None recorded

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

DATA

0 recorded samples
Source: data.csv; full acquisition settings: recipe.json; status: run.json

RUN ISSUE

RuntimeError: RP is not marked as running the Pyrpl FPGA; refusing to reload it during passive measurement

Run log

RECORDED OUTPUT

Run failed: RuntimeError: RP is not marked as running the Pyrpl FPGA; refusing to reload it during passive measurement

Active parameters

Contrast: pd, passive. Active drive readback is in drive.json; voltages are RP command volts.

Parameter	Recorded value
label	"rp-quick-test"
comment	""
duration_s	12.0
report	"pdf"
pd_input	"in1"
rp_output_max_voltage	1.0
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
visibility_frequency_hz	0.5
visibility_source	"pd"
visibility_mode	"passive"
visibility_axis	"phi1"
visibility_waveform	"sin"
visibility_amplitude_v	0.3875
visibility_offset_v	0.3875
visibility_pax_sample_period_s	0.05
visibility_dark_voltage_v	null